

Bioinformatics Analysis Report: Breaking seed dormancy in Mediterranean Brassica rapa wild populations: is cold plasma treatment efficient?

Paper Information

- **Paper ID:** 2512_23114
 - **Data Type:** Phenotypic seed germination traits, climate data, microscopy images, electrical measurements
 - **Organism:** Brassica rapa (turnip)
 - **Pages Analyzed:** 12
 - **Figures Extracted:** 11
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Analysis Pipeline

Step 1: Seed Material Production and Preparation

Description: Seeds from 61 wild B. rapa populations were collected from Mediterranean regions and produced under standardized conditions in Le Rheu, France. Seeds were stored under controlled conditions before phenotyping.

Tools: Standard seed threshing equipment, Climate-controlled storage

Input: Wild-collected seeds from at least 15 mother plants per population

Output: Standardized seed lots (~18 months old, stored at 4°C, 10% RH) ready for experiments

Parameters: - `production_location` : Le Rheu, France -
`vernalization` : 1 month at 4-6°C, 8-hour day length for 4-5 leaf-stage seedlings - `storage_conditions` : 4°C, 10% relative humidity, dark, ~18 months - `sample_size_per_population` : 1050 seeds for plasma treatment

Best Practice Note: Standardized seed production ensures uniform maternal environment effects. Storage conditions maintained seed viability while preserving dormancy characteristics.

Step 2: Climate Data Extraction

Description: Geo-referenced collection locations used to extract climate data from WorldClim database to characterize native environmental conditions.

Tools: WorldClim database, Geographic Information System (GIS) tools

Input: GPS coordinates of 61 collection sites

Output: Climate variables (mean annual precipitation, minimum/maximum temperatures)

Parameters: - `database` : WorldClim - `temporal_period` : 2010-2018 - `spatial_resolution` : 1 km accuracy - `variables_extracted` : ['MAP (mean annual precipitation)', 'TMin (minimum temperature)', 'TMax (maximum temperature)']

Best Practice Note: Climate data helps correlate dormancy traits with native habitat conditions. No specific extraction tools mentioned; QGIS or R packages (raster, terra) would be standard.

Step 3: Cold Plasma Treatment

Description: Seeds exposed to dielectric barrier discharge (DBD) cold plasma using custom-built parallel system generating ambient air plasma for dormancy breaking.

Tools: Custom 8-unit DBD plasma system, High-voltage generator (ELC Anecy GF467AF function generator + Crest Audio CC5500 power amplifier), Digital oscilloscope (Teledyne LeCroy Wavesurfer 3054, 500 MHz, 4 GS/s),

High-voltage probes (Tektronix P6015A, Teledyne LeCroy PPE 20 kV), Current transformer (Pearson model 2877)

Input: 1050 seeds per population in 8 parallel units

Output: Plasma-treated seeds ready for germination tests

Parameters: - voltage : 6 kV amplitude (sinusoidal) - frequency : 500 Hz
- duration : 10 minutes - gas_type : Ambient air - gap_distance : 1-2 mm (reported inconsistently: Figure 2 shows 2 mm, text states 1 mm) -
electrode_configuration : Plane-to-plane, 1 mm dielectric barrier -
ballast_resistor : 250 kΩ, 600 W - monitoring : Voltage (CH1, CH2) and discharge current (CH3)

Example Command:

Not applicable - manual operation of custom hardware

Best Practice Note: Plasma parameters were not optimized per population. Best practice would include systematic parameter optimization (voltage, duration, frequency) and monitoring of reactive species production.

Step 4: Automated Germination Monitoring

Description: Dynamic germination time courses captured using automated digital imaging system under controlled temperature conditions.

Tools: In-house automated germination imaging system, Temperature-controlled chamber ($20 \pm 0.5^{\circ}\text{C}$)

Input: Four replicates of 25 seeds per population (control and plasma-treated)

Output: Raw germination counts every 2 hours for 7 days, T10, MGT, final germination percentage

Parameters: - temperature : $20 \pm 0.5^{\circ}\text{C}$ - scoring_interval : Every 2 hours - duration : 7 days - replicates : 4 - seeds_per_replicate : 25 - calculation_day : Day 5 (to minimize false positives)

Best Practice Note: No specific image analysis software mentioned. Automated systems typically use custom scripts (Python with OpenCV, or R) for radicle detection. Raw image data should be archived for reproducibility.

Step 5: Manual Dormancy Monitoring (Petri Dish Assay)

Description: Comparative assessment of dormancy release mechanisms using GA3 and chemical scarification treatments.

Tools: Petri dishes (9 cm diameter), Filter paper, Growth chamber (20°C, dark)

Input: Seeds from 61 populations (control, plasma-treated, GA3-treated, scarified)

Output: Germination percentages at day 7 for each treatment

Parameters: - `temperature` : 20°C - `light_conditions` : Dark (scored under low-intensity green light) - `scoring_interval` : Every 24 hours for 7 days - `replicates` : 4 - `seeds_per_replicate` : 25 - `GA3_concentration` : 0.2% aqueous solution - `scarification_method` : 5% NaOCl commercial bleach, 10 minutes - `rinsing` : Twice with 10 mL distilled water after scarification

Best Practice Note: Manual scoring introduces potential observer bias. Blind scoring or automated image analysis is recommended. Bleach concentration and exposure time should be optimized per species.

Step 6: Greenhouse Emergence Trial

Description: Assessment of seedling emergence in greenhouse conditions to evaluate treatment effects on seedling establishment.

Tools: Greenhouse (25 ± 5°C), Standard potting soil (Faliénor substrate with Ferti), Randomized complete block design setup

Input: 100 control and 100 plasma-treated seeds per population

Output: Emergence percentage at day 15

Parameters: - temperature : $25 \pm 5^{\circ}\text{C}$ - sowing_date : 5 June 2023 - seeds_per_block : 25-30 seeds - experimental_design : Randomized complete three-block design - assessment_time : 14 days after sowing

Best Practice Note: Block design helps control for spatial heterogeneity. Emergence data should be analyzed with GLMM to account for block effects.

Step 7: Scanning Electron Microscopy (SEM)

Description: High-resolution imaging of seed coat surface morphology to assess structural changes from plasma treatment.

Tools: JEOL JSM-7610FPlus Schottky Field Emission SEM (Tokyo, Japan)

Input: Five mature seeds per population (6 populations total, control and plasma-treated)

Output: SEM images at multiple magnifications

Parameters: - populations_analyzed : A-ANAB, A-TIBE, F-HILA, F-RAUZ, I-BISA, I-MISI - seeds_per_population : 5 - magnifications : Multiple (not specified) - voltage : Not specified

Best Practice Note: No image analysis software mentioned. ImageJ/FIJI would be standard for quantitative morphometry (cell wall thickness, surface roughness). Metadata should include accelerating voltage, magnification, and detector type.

Step 8: Histological Analysis

Description: Visualization of lignin distribution in seed coat cross-sections using phloroglucinol-HCl staining.

Tools: Leica DM750 microscope, Leica ICC50-E digital camera, Hand-sectioning with razor blade

Input: Three to five mature seeds per population

Output: Stained microscope images highlighting lignified tissues

Parameters: - `populations_analyzed` : A-KADI, I-MISI, F-HILA (representative) - `staining_method` : 1% phloroglucinol solution followed by concentrated HCl - `exposure_time` : 3 minutes in phloroglucinol solution - `objectives` : 4×, 10×, and 40×

Best Practice Note: Hand-sectioning introduces variability. Microtome sectioning would provide consistent thickness. Staining intensity should be quantified using ImageJ color deconvolution.

Step 9: Statistical Analysis

Description: Statistical comparison of germination parameters across treatments and populations.

Tools: R statistical software

Input: Germination percentages, T10, MGT, emergence data

Output: Box plots, bubble charts, correlation coefficients, significance tests

Parameters: - `correlation_method` : Pearson correlation - `test` : Tukey's HSD test - `significance_threshold` : $p > 0.05$ for non-significant differences

Example Command:

Not specified in paper

Best Practice Note: Specific R packages not mentioned. Standard would include: ggplot2 for visualization, agricolae for Tukey HSD, lme4 for mixed models. Raw data and R scripts should be deposited in repository.

Databases Used

- WorldClim (climate data, 2010-2018, 1 km resolution)
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Key Methodological Findings

- Cold plasma treatment increased mean germination from 18% to 60% across 61 wild *Brassica rapa* populations
- Plasma reduced time to 10% germination (T10) from 61 to 37 hours and mean germination time (MGT) from 66 to 56 hours
- Greenhouse emergence was not significantly improved by plasma (both ~55%)
- Scarification was most effective (79% germination) but plasma showed more uniform results than GA3 (IQR: 26% vs 40%)
- Algerian and Italian populations showed deeper dormancy associated with thicker, more lignified seed coats
- French populations exhibited non-deep dormancy with thinner seed coats and reticulate-foveolate pattern
- Plasma treatment caused minimal morphological changes but likely modified surface chemistry/permeability
- Standard plasma parameters (10 min, 6 kV, 500 Hz) were not optimized and need population-specific refinement

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